

ROMAN HNATYSHYN

AI / MACHINE LEARNING DEVELOPER

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SUMMARY

First Class Honours Computing graduate specialising in machine learning and applied AI. Builds and evaluates ML and deep-learning models — neural networks (MLP), regression and decision trees — alongside LLM, RAG and NLP applications using LangChain, LangGraph, Hugging Face Transformers and the Claude and GPT APIs. Comfortable across the full ML workflow, from data cleaning and feature engineering to model training, evaluation and deployment, with a solid grounding in algorithms and production software experience at a fintech.

EDUCATION

B.Sc. (Honours) in Computing with Software Development — First Class Honours

Munster Technological University, Tralee, Ireland | Sept 2022 – May 2026 (Level 8)

- Awarded First Class Honours on a NFQ Level 8 (Honours) degree programme.
- Implemented search-tree and genetic algorithms, neural networks and machine-learning models across coursework and projects.
- Progressed from a NFQ Level 7 Diploma (Distinction, 180 credits) into the Honours programme.

TECHNICAL SKILLS

Machine Learning & AI: Machine Learning, Deep Learning, Neural Networks (MLP), NLP, LLMs, RAG, Agentic Workflows, Prompt Engineering

Algorithms & Methods: Search Tree Algorithms, Genetic Algorithms, Neural Network Algorithms, Regression, Decision Trees, Data Analytics, Data Structures & Algorithms (DSA)

ML Libraries & Tools: scikit-learn, pandas, NumPy, Matplotlib, Hugging Face Transformers (BERT, BART, RoBERTa), LangChain, LangGraph, Pydantic, Claude API, GPT API

Programming Languages: Python, SQL / PL-SQL, Java, JavaScript

Data & Storage: MongoDB, PostgreSQL, MySQL, Redis, Chroma (Vector DB)

Cloud & MLOps: AWS (S3, Lambda, EC2), Docker, CI/CD, Git

Frameworks & Testing: FastAPI, Streamlit, Node.js, pytest, JUnit

Methodologies: Agile, Kanban

PROJECTS

CVecta — AI-Powered CV Screening & Job Matching

Python, LangChain, LangGraph, RAG, Chroma, Pydantic, Streamlit, Claude API

- Built a RAG pipeline that retrieves the most relevant CVs for a job description from a Chroma vector store and returns an LLM-scored, ranked shortlist with matched and missing skills.
- Orchestrated it as an agentic LangGraph workflow with Pydantic-validated structured output and model tiering — Claude Haiku for scoring, Sonnet for the final head-to-head comparison.
- Wrote an automated evaluation harness with positive, top-k and negative test cases that runs as a regression check in CI.
- Repository: github.com/RomaHnat/CVecta

DAVE — Documents & Applications Validation Engine

Python, Hugging Face Transformers, GPT API, MongoDB, NLP

- Built an NLP engine that classifies and validates application forms and supporting documents, automating a previously manual review workflow.

- Fine-tuned and compared transformer models (BERT, BART, RoBERTa) for document classification, selecting the best-performing model for the task.
- Integrated the GPT API to recommend the exact supporting documents each application requires.
- Designed the data pipeline and MongoDB storage layer behind model inference.
- Repository: github.com/RomaHnat/DAVE-Development

Public Health Outcome Prediction from Socio-Economic Factors

Python, scikit-learn, MLP Regressor, pandas, NumPy

- Built a multi-layer perceptron (MLP) neural-network regressor to predict population health outcomes from socio-economic indicators — economic factors, armed conflict, alcohol consumption and COVID-19 restrictions.
- Integrated and cleaned multiple public datasets, with feature selection and normalisation for neural-network training.
- Tuned network architecture and hyperparameters and evaluated predictive performance against held-out data.
- Repository: github.com/RomaHnat/Public-Health-Outcome-Prediction-from-Socio-Economic-Indicators

Sleep Quality & Disorder Analysis

Python, scikit-learn, Decision Trees, Regression, Matplotlib

- Modelled sleep quality and classified sleep disorders from health and lifestyle features — blood pressure, activity level, stress, occupation, gender, BMI category, heart rate and sleep duration.
- Applied regression and decision-tree models, using the decision tree for interpretable feature-importance analysis.
- Performed data cleaning, exploratory analysis and visualisation to surface the strongest drivers of sleep quality.
- Repository: <https://github.com/RomaHnat/Sleep-Quality-Disorder-Analysis>

EXPERIENCE

Software Development Intern

Fexco, Killorglin, Ireland | Jan 2025 – Aug 2025

- Built and tested features in a payments platform processing financial transactions in Clojure, integrating the Stripe, Braintree and Worldpay REST APIs across a microservices architecture.
- Handled database work, debugging and automated testing (Cypress) within an Agile team.

Web Developer

MTU Kerry, Tralee, Ireland | May 2024 – Aug 2024

- Developed and maintained a website for MTU using a user-centric design approach, improving access to information and resources for students, faculty and visitors
- Collaborated with stakeholders to understand business requirements and translate them into clear technical specifications and actionable feature requirements for the development team.

CERTIFICATIONS

- EDGE Bronze, Silver & Gold Certificates
- Samsung IT School Certificate

LANGUAGES

Ukrainian / Russian — Native | English — C1 Advanced | Polish — B2 Upper Intermediate